DRAFT STANDARD OPERATING PROCEDURE

Controlled Sugarcane Pollination and Hybridization

Sugarcane Breeding and Germplasm Conservation Program

SOP No.	SCB-BR-SOP-06	**Revision**	0
Effective date	_____	**Supersedes**	New
Prepared by	_____	**Reviewed by**	_____
Approved by	_____	**Controlled copy**	Yes / No

Status: Draft technical SOP. Field dimensions, thresholds, chemical treatments, quarantine steps, and local compliance requirements must be validated and approved by the implementing institution before adoption.

1. Purpose

To produce true sugarcane seed (fuzz) of known pedigree through controlled hybridization while maximizing seed set, minimizing unintended pollen contamination, and preserving complete cross records from parent selection through seed storage and sowing.

2. Scope

Covers pre-cross planning, flowering readiness, pollen testing, male/female assignment, controlled lantern/bag crossing, cross maintenance, seed maturation, fuzz harvest, seed quality testing, storage, and handoff to the seedling nursery.

3. Responsibilities

- Breeder: approves cross objectives, parent combinations, substitutes, and cross disposition.
- Crossing technician: verifies parent identity, conducts pollen tests, assembles crosses, maintains tags, and records daily observations.
- Field/flowering staff: monitors parent blocks and cuts/handles arrows without mixing.
- Seed laboratory/nursery staff: receives cross-specific fuzz, performs germination testing, and maintains cross identity during sowing.

4. Pre-cross planning

Prepare a prioritized crossing matrix before peak flowering. For every proposed mating record the female, male, breeding objective, priority, pedigree relationship, known disease/quality information, previous progeny performance, and at least one backup male or alternative female where possible.

5. Flower readiness and pollen testing

1. Inspect candidate arrows for correct parent identity, health, stage of anthesis, and physical damage.
2. Collect fresh dehiscing anthers for a pollen viability/fertility test using the station's validated staining method and microscope scoring procedure.
3. Assign provisional functional sex for that crossing session based on observed pollen fertility. Do not rely only on historical sex labels because viability can vary with environment.
4. Record stain method, number of microscope fields or grains assessed, estimated percent viable pollen, analyst, date, and parent ID.
5. Operational reference only: Sugar Research Australia reports classifying >20% viable pollen as male in its system. A local program should validate its own threshold and environmental conditions before adopting this value.

6. Preparation of female and male arrows

- Verify physical tag against the crossing database before cutting or assembly.
- Inspect the selected female immediately before enclosure. Remove already opened florets where the local validated procedure requires this to reduce the risk of prior field pollination.
- Select males with active pollen shedding and sufficient flower quantity to support the intended cross.
- Maintain cut arrows in the institution's validated crossing/holding solution. The preparation of that solution should be controlled under a separate chemical SOP.
- Assign an individual arrow number when multiple arrows from the same parent are used.

7. Controlled crossing procedure

6. Create the cross record before assembly: cross number, female, male, arrow numbers, date, breeder/technician, and breeding objective.
7. Place female and male flowers together in a clean crossing lantern/bag system. Arrange male inflorescences above the female inflorescences where the system is designed for gravity-assisted pollen movement.
8. Close and secure the enclosure to prevent unwanted pollen entry and cross mixing. Attach one external and one protected/internal label.
9. Maintain arrows hydrated in the validated solution and inspect the enclosure daily for damage, leakage, wilting, mold, or label loss.
10. During active pollen shed, gently tap/agitate male flowers only if this is part of the station's validated practice, taking care not to tear the enclosure or mix arrows.

11. Keep each cross enclosed through the principal pollination period. Sugar Research Australia reports approximately two weeks in lanterns in its operational system; local programs should adjust based on flowering duration and environment.
12. At completion of pollination, remove/discard male arrows according to station practice while keeping females alive and protected for seed maturation.
13. Allow female inflorescences to mature until fuzz is ready for harvest. Record abnormalities, failed crosses, or suspected contamination immediately.

8. Labeling standard

Every cross must have at least two independent labels. Minimum information: cross number; female parent; male parent; date assembled; female and male arrow numbers; technician initials. A barcode linked to the digital cross record is strongly recommended but must supplement, not replace, human-readable text.

9. Fuzz harvest and seed handling

14. Harvest each female/cross separately into a clean, breathable, pre-labeled bag. Never combine crosses unless intentionally creating a documented polycross.
15. Verify cross label against the cross register before the bag leaves the crossing area.
16. Dry fuzz under the institution's validated low-humidity conditions. SRA Australia describes a controlled cool/dehumidified drying step before seed evaluation; the exact temperature/humidity should be locally validated.
17. Record gross fuzz/seed weight and, where enough material is available, perform a representative germination test.
18. Classify seed lot quality and decide whether to retain, resow, or discard according to program thresholds.
19. Package and store seed under validated conditions with cross number, harvest date, weight, germination result, packet/container ID, and storage location.
20. At sowing, maintain identity as Cross -> seed packet -> seed tray/box -> family tray -> field population.

10. Cross failure and contamination handling

- Failed pollination or poor seed set: record the outcome; do not silently reuse the cross number for a different mating.
- Suspected pollen contamination: quarantine the seed lot from controlled-pedigree use; mark status in database and decide whether it may be used only as open/uncertain pedigree material.
- Label loss: stop processing; reconstruct identity only if an unambiguous independent record exists. Otherwise classify as unknown rather than guessing.
- Mold/wilting or broken enclosure: document event and evaluate whether seed identity or viability remains acceptable.

11. Quality-control checkpoints

- Parent ID verified immediately before cross assembly.
- Current-session pollen result recorded before male/female assignment.
- Two labels per cross; digital record created before or at assembly.
- No cross harvested without matching register entry.
- Seed packet, germination test, and sowing tray retain the same cross number.
- Cross-number corrections preserve an audit trail and never overwrite original pedigree without authorization.

12. Cross register and pollination record

Cross no.	Female	Male	Pollen %	Date set	Arrow nos.	Date male removed	Harvest date	Fuzz wt.	Germination	Status

Cross no.	Female	Male	Pollen %	Date set	Arrow nos.	Date male removed	Harvest date	Fuzz wt.	Germination	Status

References and technical basis

Sugar Research Australia - detailed controlled crossing process, pollen testing, lantern crossing, seed maturation and storage - Web source

Sugar Regulatory Administration - research portfolio including Pollination, Sowing, and Seedling Care - Web source